

Nickolas Yanez

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EDUCATION

The University of Texas at Austin

B.S. Aerospace Engineering Honors, Minor in Robotics

- **GPA:** 3.82/4.00

Austin, TX

May 2027

EXPERIENCE

Hardware Team Lead — Active Roll Control

Apr. 2026 – Present

Longhorn Rocketry Association - Advanced Research

Austin, TX

- Lead a 6-member hardware team: run design meetings, make design decisions, and set next-step priorities against a fixed build deadline
- Led redesign from fin-tab roll control to four forward canards after fin-embedded servos created packaging, reusability, and wiring problems
- Designed a fully 3D-printed ABS upper body in Fusion 360 with an integrated servo casing/e-bay and four removable access panels; tuned fits and clearances through iterative print tests
- Modeled the 31.25-in vehicle in OpenRocket at 1.16-cal stability and verified servo torque against a $3\times$ design hinge load; found the canard pivot aft of the aerodynamic center and directed a relocation that cuts hinge torque $\sim 7\times$

Mechanical Member — Air Brakes

Aug. 2025 – Apr. 2026

Longhorn Rocketry Association - Advanced Research

Austin, TX

- Researched flight-proven collegiate air brake designs and pushed the team to adopt a cam-slot flap mechanism, selected at PDR for its simplicity, low weight, and low profile
- Helped define the expected aerodynamic load case for apogee reduction from 12,000 ft to 10,000 ft on a 6-in competition rocket, and planned integration into the competition airframe

Vehicle Integration Member - Flight Hardware

Aug. 2024 – Aug. 2025

Longhorn Rocketry Association - Flight Systems

Austin, TX

- Packed parachutes and rigged shock cords; supported launch-readiness checks for test and competition flights
- Fabricated and integrated composite airframes, fins, and bonded structures
- Verified a 2.75-cal stability margin in OpenRocket for a 134-in, Mach 1.87 competition vehicle
- Selected as 1 of 17 members to support flight operations at the 2025 Spaceport America Cup

PROJECTS

Packaging Drop Test | *Electromechanical Systems*

Fall 2025

- Designed and wired the instrumentation: a ± 500 g piezoelectric accelerometer and a photodiode-triggered LabVIEW DAQ sampling at 10 kHz to capture drop-impact shock
- Built the drop rig and pushed to double planned trials ($5 \rightarrow 10$ per configuration) to reduce random uncertainty, for 40 total drops from 15 in
- Helped analyze impact pulses; the best cushioning cut peak payload deceleration from ~ 580 to ~ 220 m/s² ($\sim 62\%$)

3-DOF Robotic Leg | *Gateway to Robotics*

Fall 2025

- Fabricated a 3-DOF robotic leg using 3D-printed parts, laser-cut hardware, and Dynamixel servos
- Implemented forward and inverse kinematics in Python using homogeneous transformations and Newton's method
- Integrated hardware with PyBullet for bidirectional simulation control and experimentally evaluated PD response

Quadrotor Dynamics Simulation & Control | *Aerial Robotics*

Spring 2026

- Built a MATLAB quadrotor simulation with 6-DOF dynamics, motor modeling, and aerodynamic drag
- Implemented closed-loop position and attitude control with simulated GNSS/IMU sensing and UKF state estimation
- Implemented stereo 3D localization in OpenCV using RANSAC outlier rejection and least-squares triangulation

TECHNICAL SKILLS

Programming: MATLAB, Python, C++ | **Tools:** SolidWorks, Fusion 360, OpenRocket, LabVIEW, ROS, OpenCV, Git, Linux

Hardware: 3D printing, composite fabrication, DAQ instrumentation, flight vehicle integration

Coursework: Structural Dynamics, Electromechanical Systems, Launch Vehicle Design, Propulsion, Feedback Control

Honors: Hans Mark Engineering Honors Scholar, CH Foundation Scholar